

AKRON-CANTON, OH, USA

WMO: 725210

Lat: 40.918N

Lon: 81.444W

Elev: 1213

StdP: 14.06

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.2	(c) 7.3	(d) -5.4	(e) 4.3	(f) 3.6	(g) -0.7	(h) 5.5	(i) 9.1	(j) 26.6	(k) 33.0	(l) 24.6	(m) 30.5	(n) 9.1	(o) 250	(p) 0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	88.8	72.8	86.3	71.6	84.0	70.2	75.4	84.8	73.9	82.5	72.5	80.4	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	126.2	80.6	71.1	119.9	78.4	69.7	114.3	77.1	39.7	85.1	38.2	82.7	36.8	80.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.4	(b) 20.0	(c) 18.1	(e) -3.5	(f) 92.0	(g) 6.7	(h) 2.9	(i) -8.2	(j) 94.0	(k) -12.1	(l) 95.7	(m) -15.9	(n) 97.3	(o) -20.7	(p) 99.4		
			DB	-4.1	78.9	6.5	2.7	-8.8	80.8	-12.6	82.4	-16.3	83.9	-21.0	85.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	26.6	29.3	37.8	50.0	60.4	69.1	72.7	71.4	64.8	53.1	41.6	32.2
	DBStd	18.61	12.17	11.38	11.28	9.53	8.55	6.59	5.21	5.14	7.20	8.72	9.42	10.18
	HDD50	2757	730	585	408	116	12	0	0	0	1	67	282	556
	HDD65	5923	1191	1000	843	455	196	37	4	9	90	381	701	1016
	CDD50	3078	4	5	31	117	335	573	704	663	445	163	32	6
	CDD65	768	0	0	1	6	54	160	244	207	84	12	0	0
	CDH74	5840	0	0	10	68	445	1205	1931	1487	619	74	1	0
	CDH80	1631	0	0	0	5	90	343	603	410	169	11	0	0
(14) Wind	WSAvg	8.9	10.8	10.3	10.1	10.1	8.5	7.7	7.0	6.7	7.2	8.5	9.5	10.0
(15) Precipitation	PrecAvg	38.9	2.5	2.3	3.2	3.5	3.9	3.8	4.0	3.5	3.4	2.8	3.0	2.9
	PrecMax	65.7	6.1	5.7	8.8	6.5	8.1	9.3	12.6	8.2	9.0	7.1	9.4	6.7
	PrecMin	23.8	0.7	0.3	1.3	0.9	1.1	0.4	0.7	0.5	0.6	0.5	0.6	0.9
	PrecStd	7.3	1.2	1.1	1.3	1.4	1.8	2.1	2.4	1.9	1.8	1.4	1.5	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	64.8	73.9	80.3	86.2	89.9	91.9	90.4	89.8	81.1	69.3	61.6
		MCWB	55.1	56.0	59.2	63.2	69.2	72.9	75.9	73.2	71.4	68.4	58.6	56.0
	2%	DB	56.0	57.0	66.8	75.6	82.5	87.0	88.8	87.4	84.9	75.4	64.4	56.6
		MCWB	52.6	51.3	55.2	60.2	67.3	71.6	73.9	71.8	70.3	64.7	56.6	52.5
	5%	DB	50.4	51.1	61.1	70.9	79.3	84.2	86.1	84.6	80.7	71.4	60.5	52.2
		MCWB	46.7	45.9	52.9	57.8	65.9	70.4	72.1	70.8	68.4	62.2	53.8	48.4
	10%	DB	44.0	45.4	55.9	66.4	75.5	81.1	83.4	81.8	77.1	67.4	56.5	47.1
		MCWB	40.3	41.0	49.0	55.0	64.0	68.5	70.3	69.7	66.5	59.9	51.0	43.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.7	57.5	60.5	66.1	72.3	76.1	78.6	77.0	75.0	70.8	62.0	58.0
		MCDB	59.4	63.8	71.1	76.1	82.4	86.1	87.9	85.8	85.4	77.8	67.3	60.4
	2%	WB	53.4	52.0	57.2	62.6	69.8	73.8	76.0	74.7	72.4	67.1	58.2	53.4
		MCDB	56.1	55.6	64.2	71.8	79.2	83.3	85.3	83.3	80.8	72.7	62.6	56.2
	5%	WB	47.2	46.2	53.8	59.9	67.8	72.2	74.2	73.1	70.3	63.6	54.9	48.6
		MCDB	49.7	50.6	60.0	68.0	76.5	81.0	82.8	80.5	77.8	69.3	59.2	51.3
	10%	WB	40.6	41.3	49.6	57.0	65.6	70.5	72.5	71.5	68.4	60.4	51.4	43.6
		MCDB	43.7	45.0	55.7	64.3	73.5	78.3	80.3	78.3	75.0	66.3	56.0	47.2
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	14.8	17.1	20.0	19.5	18.6	18.5	18.2	18.6	17.2	14.8	12.2
		MCDBR	18.3	21.3	24.2	25.1	22.8	21.1	20.3	20.4	21.6	21.3	20.3	17.7
		MCWBR	16.0	17.2	15.6	15.0	11.7	10.0	8.9	8.8	9.9	12.3	14.7	14.9
	5% WB	MCDBR	17.6	20.2	21.7	21.8	20.0	18.7	18.2	17.4	18.3	18.3	18.3	17.2
		MCWBR	16.3	17.5	15.2	14.8	11.6	10.0	9.0	8.6	9.5	12.2	15.3	16.1
(40) Clear-Sky Solar Irradiance	taub		0.299	0.324	0.352	0.385	0.423	0.446	0.447	0.436	0.381	0.358	0.315	0.301
	taud		2.434	2.347	2.351	2.291	2.209	2.182	2.196	2.218	2.347	2.405	2.501	2.467
	Ebn at Noon		273	280	285	281	272	265	263	263	271	265	265	262
	Edh at Noon		25	32	36	40	45	46	45	43	35	30	23	23
(44) All-Sky Solar Radiation	RadAvg		458	735	1043	1423	1699	1861	1858	1679	1350	884	575	383
	RadStd		40	95	106	137	143	143	128	86	123	80	85	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.37	N/A	N/A	+1.30	N/A	N/A	N/A	-326	+322	+185
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+113

Nomenclature: See separate page